

Weights or Skills?

A Survey of Robot-Learning Techniques—from Action-Predicting Weights to Robots that Write their Own Skills

GAYTRI JENA, UC Berkeley, USA

KAPIL WANASKAR, Canva Research, USA

VINIJA JAIN, Google, USA

AMAN CHADHA, Google DeepMind, USA

VASU SHARMA, PocketFM, USA

AMITAVA DAS, Pragya Lab, BITS Pilani Goa, India

Robot learning is splitting into two bets: policies that bake competence into *frozen weights*—vision-language-action (VLA) models—and agents that write and refine their own *executable skills* as code. This survey organises the field around that axis, **weights or skills**, and makes its central contribution a deep-dive that arranges code-as-policy methods by their *degree of self-improvement*: from zero-shot program synthesis, through closed-loop self-repair and persistent skill memory, to the near-empty cell where execution feedback, skill memory, and evolutionary search combine into one open-ended loop. We anchor that frontier on ASPIRE and its real-world sibling ENPIRE, and we map the complementary “skills” pole—from unsupervised reinforcement-learning skill discovery to large-language-model skill libraries—showing that skills come in an RL flavour and a code flavour, of which only the code flavour self-improves without gradients. Finally, we connect the taxonomy to the emerging *skill economy*: commercial robot-skill marketplaces now distribute one-tap skills across robots, but ship only static playback, and we lay out the open problems—adaptation, cross-embodiment portability, provenance, safety verification, composition, and standardisation—that separate a store of animations from a store of capabilities. The survey covers roughly seventy-five systems across six technique families and provides a canonical taxonomy figure for each.

CCS Concepts: • **Computing methodologies** → **Robotic planning**; *Machine learning*.

Additional Key Words and Phrases: robot learning, code-as-policy, self-improvement, vision-language-action models, skill libraries, physical AI, survey

1 INTRODUCTION

Two paradigms now dominate robot learning. *Vision-language-action* (VLA) models predict actions from frozen weights; *code-as-policy* agents instead write executable programs and improve them from experience [32, 39]. This survey is organized around that fault line.

Why now: the skill stack. The app store for robot skills has arrived—Unitree UniStore ships one-tap, cross-model motion downloads [55]. But every skill in it is *canned playback*: the least-capable point on our map. The skill stack has three layers—**(1) Build** (how skills are created, §2–§8), **(2) Distribute** (marketplaces and hubs), and **(3) Adapt** (on-device self-improvement, §3). UniStore is layer 2 over static layer-1 skills; the promised value—“adjust grip per fruit so nothing bruises”—lives in layer 3, which no shipped skill does today.

Contributions. (i) a taxonomy of robot-learning techniques organized by *weights vs. skills* (§2); (ii) a deep-dive arranging code-as-policy by *degree of self-improvement* and isolating the un-surveyed §3.1.5 cell (§3); (iii) a companion map of the “skills” pole (§6); and (iv) an open-problems agenda for the emerging skill economy (§9).

Authors’ addresses: Gaytri Jena, UC Berkeley, Berkeley, CA, USA; Kapil Wanaskar, Canva Research, USA; Vinija Jain, Google, USA; Aman Chadha, Google DeepMind, USA; Vasu Sharma, PocketFM, USA; Amitava Das, Pragya Lab, BITS Pilani Goa, India.

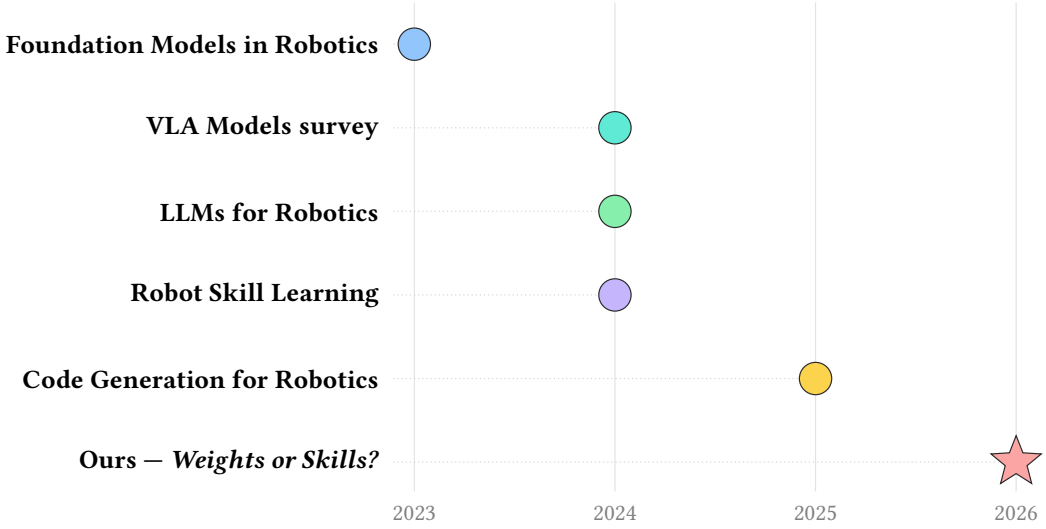


Table 1. Related surveys on a timeline. Prior surveys (2023–2025) each cover part of the landscape—foundation models, VLA policies, LLM planning, skill acquisition, code generation—but none organizes code-as-policy by *degree of self-improvement*. **Ours** (★, 2026) isolates the feedback + memory + search cell (§3.1.5) that ASPIRE occupies.

Related surveys. Table 1 places the closest prior surveys on a timeline. Each covers part of the landscape—foundation models, VLA policies, LLM-based planning, skill acquisition, or code generation—but none organises code-as-policy by *degree of self-improvement* or isolates the §3.1.5 cell, which is the gap this survey fills.

2 A TAXONOMY OF ROBOT-LEARNING TECHNIQUES

Figure 1 organizes the field into six branches. The dividing question is *what ships*: frozen weights (§4) or executable skills (§3). We tag mechanisms by **Feedback**, **Search**, and **Memory** throughout.

3 CODE-AS-POLICY: ROBOTS THAT WRITE THEIR OWN SKILLS

The organising contribution of this survey is to arrange code-as-policy methods not by application or robot embodiment, but by *how much of their own competence a system produces at run time*. Concretely, we ask five questions in sequence: does the agent (i) write control code at all, (ii) repair that code from execution feedback within a task, (iii) remember validated code across tasks, (iv) search a population of programs rather than following a single trajectory of repairs, and (v) close all of these into one open-ended loop? Each “yes” is a rung on the ladder of Figure 2, and the population thins sharply toward the top: of the thirteen zero-shot systems, only a handful ever reach the combined feedback–memory–search regime. The novelty of the survey is naming this progression and showing that the top rung is nearly empty.

3.1 Zero-shot synthesis (§3.1.1)

The foundational move is to treat the language model as a *program synthesiser* over a fixed library of perception and control primitives. *Code-as-Policies* [32] showed that an LLM prompted with an API and a natural-language instruction can emit executable Python that composes those primitives, recursively defining undefined functions and parameterising control with arithmetic and feedback

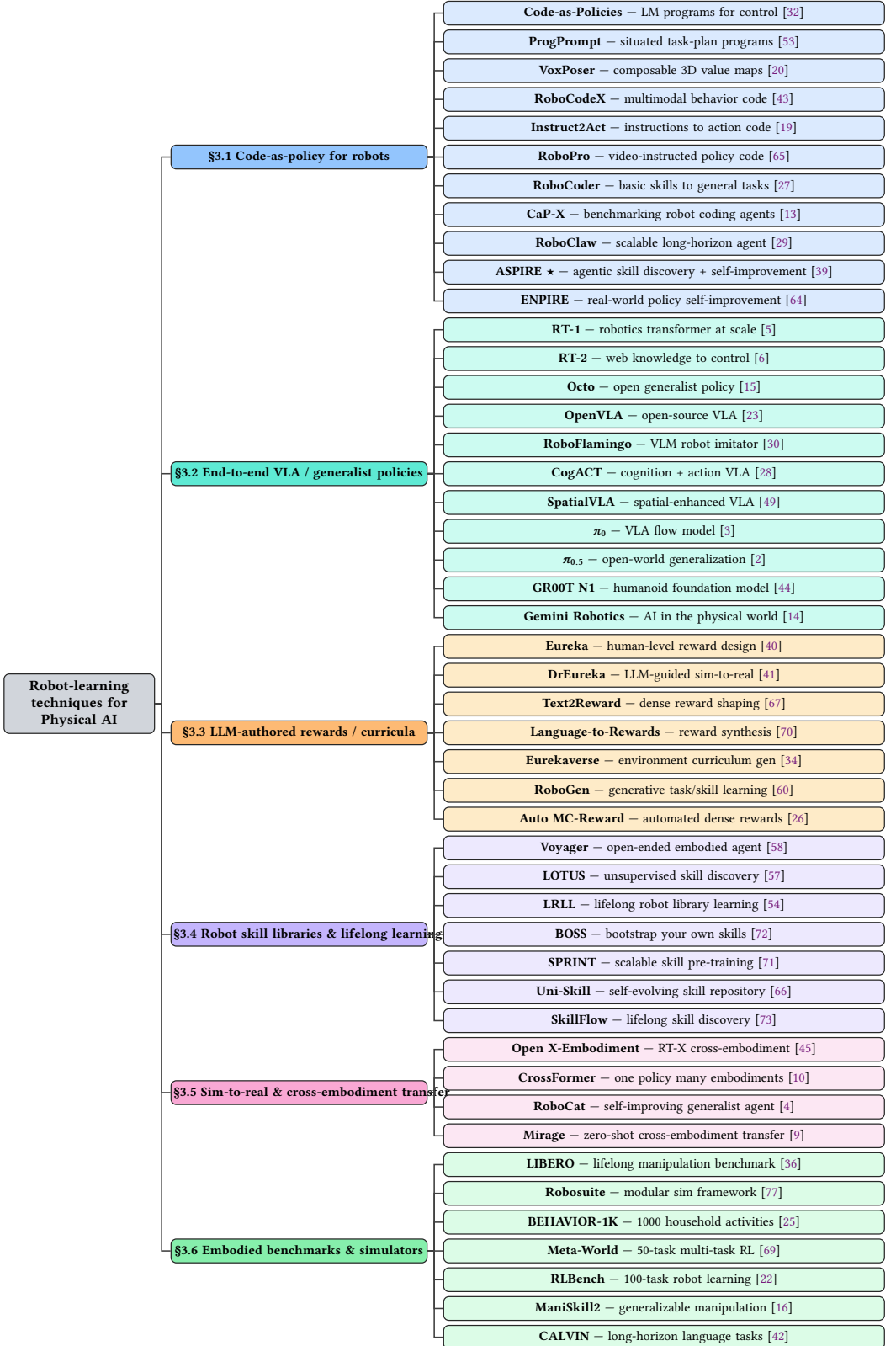


Fig. 1. Taxonomy of robot-learning techniques for Physical AI (~47 systems). The fault line: §3.1 ships *code* and §3.2 ships *weights*. **ASPIRE ★** anchors the self-improving code-as-policy frontier (§3).

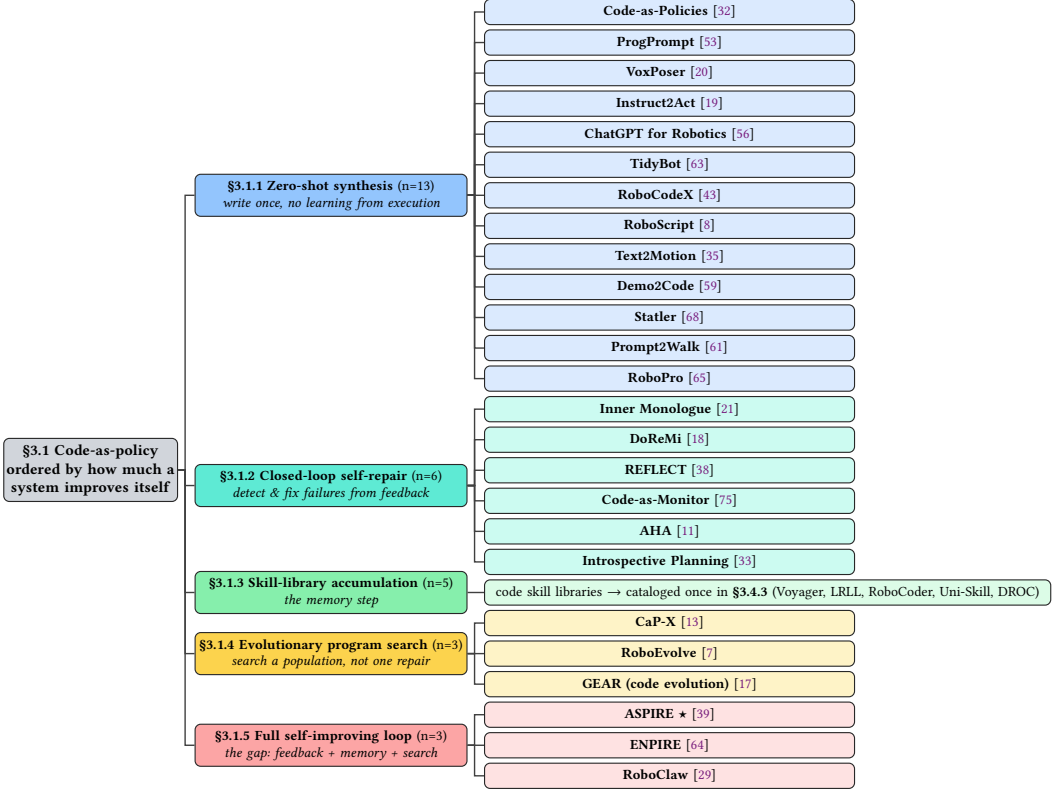


Fig. 2. The self-improvement funnel within code-as-policy ($13 \rightarrow 6 \rightarrow 5 \rightarrow 3 \rightarrow 3$), one colour per sub-section. As the demand rises (write once $\rightarrow$ repair $\rightarrow$ remember $\rightarrow$ search $\rightarrow$ all three), occupants thin out; §3.1.5—feedback + memory + search combined—is nearly empty, and only **ASPIRE ★** was built for it end-to-end.

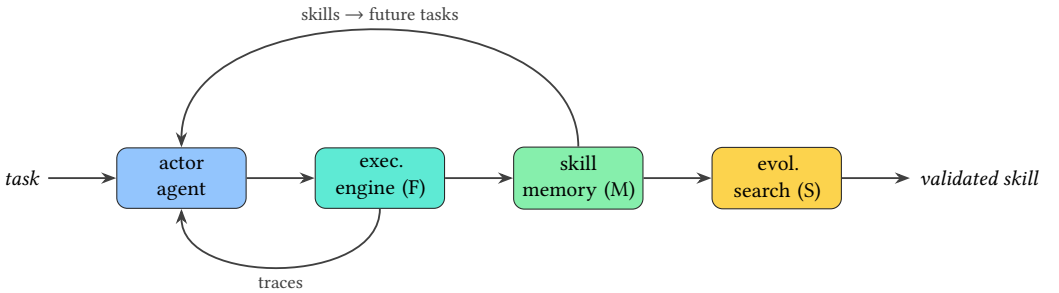


Fig. 3. The full self-improving loop (§3.1.5): feedback (F) + memory (M) + search (S) combine into one open-ended loop—the ASPIRE cell. VLAs (§3.2) have no such loop.

logic. *ProgPrompt* [53] situates the generated program in the scene by exposing available actions and objects as importable symbols, while *VoxPoser* [20] has the model write code that *composes 3D value maps* for a motion planner rather than calling fixed skills, loosening the dependence on a hand-authored API. Subsequent work broadened the input modality and task horizon—*RoboCodeX* [43]

conditions on multimodal observations, *Instruct2Act* [19] maps instructions to perception–action code, and video- and demonstration-conditioned variants such as *RoboPro* [65], *Demo2Code* [59] and *Text2Motion* [35] recover programs from richer context. *Statler* [68] maintains an explicit world state across steps, and *TidyBot* [63] and *ChatGPT for Robotics* [56] demonstrate personalisation and prompt design for real hardware. What unites this rung, and limits it, is that the program is written *once*: there is no channel from execution back to the synthesiser, so a plan that fails at run time simply fails.

3.2 Closed-loop self-repair (§3.1.2)

The second rung adds a within-task feedback loop. *Inner Monologue* [21] feeds success detectors, scene descriptions and human feedback back into the model as language, letting it replan when a step fails. *DoReMi* [18] detects misalignments between plan and execution and triggers recovery, and *REFLECT* [38] builds a hierarchical, multimodal summary of past interaction that an LLM queries to explain a failure and propose a correction. More recent systems push the monitor into the perception loop—*Code-as-Monitor* [75] compiles spatio-temporal constraints into vision-language checks, and *AHA* [11] trains a model to reason explicitly about failure modes. The novelty of this rung is the closed loop, but it is a *short* loop: corrections are discarded at task boundaries, so nothing accumulates.

3.3 Skill-library accumulation (§3.1.3)

The third rung makes the loop persistent by writing validated code into a growing library that later tasks retrieve and compose. Because this “memory” axis is a full technique family in its own right, we catalogue its systems once, in §6 (§3.4.3)—*Voyager* [58] is the canonical example, curating an ever-growing library of verified skill programs with when-to-apply guards. The key distinction from §3.1.2 is temporal scope: competence compounds *across* tasks rather than being rebuilt each time.

3.4 Evolutionary program search (§3.1.4)

The fourth rung replaces a single chain of repairs with population-based search over programs. Rather than iteratively debugging one candidate, the agent maintains several, scores them by execution, and mutates the best. *CaP-X* [13] provides an interactive gym and benchmark for exactly this style of coding agent, and *RoboEvolve* [7] couples a planner and a learned simulator into a co-evolutionary loop that mines near-miss failures to stabilise search. The novelty is the shift from *repair* to *search*: exploration is explicit and parallel, which trades compute for a better chance of escaping local failures.

3.5 The full self-improving loop (§3.1.5)

The top rung combines all three mechanisms—execution feedback (F), a persistent skill memory (M), and evolutionary search (S)—into one open-ended loop (Figure 3). *ASPIRE* [39] instantiates this with a robot execution engine that emits per-primitive multimodal traces, a skill library of validated code with guards, and a population search over candidate programs, so that experience compounds and transfers across tasks and embodiments. Its real-world sibling *ENPIRE* [64] closes the same loop on physical hardware, using automatic reset, parallel rollouts, and log-driven algorithm revision to drive a coding agent to near-perfect success on dexterous tasks. This cell—feedback *and* memory *and* search—is what no prior survey isolates, and it is nearly empty: it is the flag this survey plants.

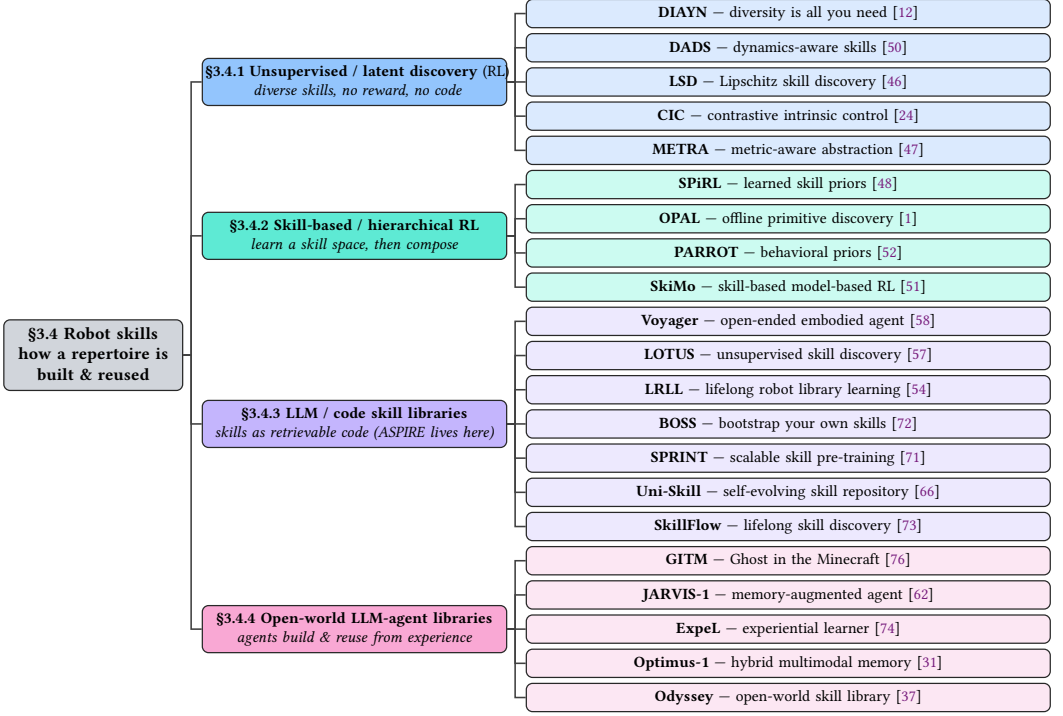


Fig. 4. The “skills” pole, one colour per sub-section. §3.4.1–2 build skills as *RL policies* (mature, included as landscape); §3.4.3–4 build them as *retrievable code*, where ASPIRE’s self-improving §3.1 lives.

4 END-TO-END VISION-LANGUAGE-ACTION MODELS

VLA models bake competence into frozen weights [3, 23]: image and language in, action chunks out, no loop and no code. They are the foil against which the self-improving code camp (§3) is defined.

5 REWARD AND CURRICULUM SYNTHESIS

Here the LLM writes *reward* code rather than policy code; an RL loop trains on it and a reflection step revises it [40]. One feedback loop, but the artifact shipped is still weights, not a reusable skill.

6 SKILL LIBRARIES: HOW A REPERTOIRE IS BUILT

Figure 4 maps how a skill repertoire is built: unsupervised/latent discovery in RL [12], skill-based/hierarchical RL [48], LLM/code skill libraries [58], and open-world LLM-agent libraries. Skills come in an *RL flavor* (latent policies, gradient-trained) and a *code flavor* (retrievable programs); only the code flavor self-improves without gradients, linking back to §3.

7 SIM-TO-REAL AND CROSS-EMBODIMENT TRANSFER

A skill is only useful if it moves across robots. Cross-embodiment datasets and shared action spaces train one policy for many bodies [45]; UniStore’s “one skill, G1/H1/B2/Go2” claim (§9) is this problem in production.

Dimension	ASPIRE [39] <i>code+skills</i>	π_0 [3] <i>VLA weights</i>	Code-as-Pol. [32] <i>code, no mem.</i>	Eureka [40] <i>reward</i>
What it outputs	skill code+lib	action chunks	policy code	reward code
How it improves	F+M+S loop	gradient pretrain	– one-shot	RL+reflection
Gradient training?	✗	✓	✗	✓
Persistent skill library?	✓	✗	✗	✗
Failure feedback signal	exec. traces	–	–	train. stats
Search strategy	evolutionary	–	–	–
Real-robot validation	✓	✓	✓	✓
Continual / open-ended?	✓	✗	✗	✗
Beyond fixed APIs?	✓	✓	✗	–
Interpretable / editable?	✓	✗	✓	✓

Table 2. Capability matrix, one exemplar per camp; each method header links to its entry in the References. Only ASPIRE combines a persistent skill library with feedback- and search-driven self-improvement. ✓ = yes, ✗ = no, – = n/a.

8 BENCHMARKS AND EVALUATION

Table 2 contrasts one exemplar per camp. Standard suites—LIBERO, CALVIN, Meta-World, RLBench, ManiSkill2, BEHAVIOR-1K [16, 22, 25, 36, 42, 69]—report success rate and completed-task metrics, but few probe *self-improvement over time*.

9 THE SKILL ECONOMY: OPEN PROBLEMS

The arrival of a commercial marketplace for robot skills—Unitree’s UniStore ships one-tap, cross-model motion packages [55]—turns several research questions from hypothetical into pressing. We frame them as the open agenda of this survey.

The static-to-adaptive gap. Every skill shipped today is, in effect, replayed: a motion package or a fixed program is installed and executed without perceiving whether *this* kitchen, gripper, or object differs from the one it was authored for. The techniques of §3—within-task repair, persistent memory, and search—are precisely what would let a downloaded skill *adapt* on the target robot. Closing this gap is the difference between a store of animations and a store of capabilities, and it is the direct application of the §3.1.5 loop to a distribution setting where the base skill is given rather than discovered.

Cross-embodiment portability. UniStore advertises a single skill running across the G1 and H1 humanoids and the B2/Go2 quadrupeds—bodies with different morphologies, action spaces, and dynamics. This is the cross-embodiment transfer problem (§7) at deployment scale: without a shared action interface [45] or explicit embodiment adaptation, a package tuned for one platform has no guarantee of correctness on another. Formal portability—what must hold for a skill to be certified on a new body—remains open.

Provenance and trust. A marketplace invites third-party, crowdsourced uploads, including raw motion-capture data and code from unknown authors. Robot skills act on the physical world, so provenance (who authored a skill, from what data, validated how) becomes a safety property, not merely a metadata nicety. There is no accepted standard for signing, attesting, or auditing a robot skill’s origin.

Safety verification. Vendors promise that “every skill is scanned, tested, and verified,” but for a code-as-policy skill this is undecidable in general and hard in practice: the skill’s effect depends on the scene it meets. Practical questions include which pre-conditions a skill must declare, what sandboxed or simulated screening it must pass, and how run-time monitors (cf. *Code-as-Monitor* [75]) can bound behaviour on unseen inputs.

Skill composition. Downloading two skills should ideally yield a third: “make coffee” plus “clear the table” composed into a morning routine. Composition requires shared interfaces and a calculus over pre-/post-conditions; today’s motion packages are opaque and do not compose, and code skills compose only within a single authoring system.

Portability standards and skill ontologies. Reuse at scale needs a shared vocabulary: declared pre-conditions, expected effects, and embodiment-capability profiles, with relations such as *requires*, *provides*, and *substitutable-by*. Prior work on skill ontologies and hardware-level reusability points the way, but no cross-vendor standard exists, which is what a genuine ecosystem would require.

Local versus shared libraries. Finally, the field must reconcile two notions of “skill library” that this survey has kept distinct: the *self-built* library a robot grows from its own experience (ASPIRE, §3) and the *shared* library a robot downloads from a marketplace (UniStore). The interesting regime is the hybrid—an agent that both contributes to and draws from a commons—and its incentives, quality control, and feedback dynamics are entirely unstudied.

10 LIMITATIONS OF THIS SURVEY

We state the boundaries of the survey explicitly. First, our organising axis—*degree of self-improvement*—is deliberately code-centric; it foregrounds systems that emit and revise programs and treats weight-based policies (§4) and reinforcement-learning skill discovery (§6) as context rather than as the object of the taxonomy. A survey centred on representation learning or on real-world data collection would draw the map differently. Second, the frontier we emphasise (§3.1.5) is populated by very recent, in some cases concurrent, systems; benchmark numbers across them are not directly comparable, and we therefore report capabilities qualitatively rather than ranking methods. Third, the field moves quickly: several 2025–2026 systems we cite were released while this survey was being written, and the coverage should be read as a snapshot. Finally, the “skill economy” framing (§9) is grounded in an emerging commercial trend (one vendor’s marketplace at the time of writing); we treat it as motivation for open problems, not as a mature body of results.

11 FUTURE DIRECTIONS

Three directions follow directly from the map. **(1) Standardised evaluation of self-improvement.** The systems on the top rung (§3.1.5) each report improvement on their own splits; the field lacks a shared protocol that measures held-out success as a function of accumulated experience (an area-under-the-learning-curve metric), cross-embodiment transfer, and library-reuse rate under a common budget. Building such a benchmark is the natural companion to this survey. **(2) Adaptation as a first-class marketplace primitive.** The open problems of §9 suggest a research programme in which a downloaded skill is not a frozen artefact but a starting point that the §3 loop specialises to the local robot and environment—turning distribution (layer 2) and adaptation (layer 3) into a single pipeline. **(3) Bridging the RL and code flavours of skills.** §6 shows that skills are learned either as latent RL policies or as retrievable code; a unifying interface—code that wraps, calls, and refines learned policies, and learned policies distilled back into inspectable code—would let the self-improvement machinery of §3 operate over both. We see the combination of a standard

self-improvement benchmark with an adaptation-centric marketplace as the most likely path from today’s static skill stores to genuinely capable, compounding robots.

12 CONCLUSION

Robot learning is organising itself around a single question: should competence be shipped as *weights* or as *skills*? This survey took that question as its axis. We mapped the field into six branches (§2), positioned end-to-end vision-language- action models as the weights-based foil (§4), and gave the code-as-policy camp its own deep-dive (§3) organised by *degree of self-improvement*: from one-shot program synthesis, through within-task repair, persistent skill memory, and population search, to the open-ended loop that combines all three. That top cell—feedback + memory + search—is the least populated and least surveyed region of the field, and it is where systems such as ASPIRE and ENPIRE now sit. We complemented this with a map of how skill *repertoires* are built (§6), showing that skills come in an RL flavour and a code flavour of which only the latter self-improves without gradients, and we connected the academic taxonomy to the nascent skill economy (§9), where commercial marketplaces already distribute skills but ship only static playback. The gap between that static distribution and genuine on-device adaptation is, we argue, the defining research opportunity for physical AI, and the self-improvement ladder of §3 is the road across it.

REFERENCES

- [1] Anurag Ajay, Aviral Kumar, Pulkit Agrawal, Sergey Levine, and Ofir Nachum. 2021. OPAL: Offline Primitive Discovery for Accelerating Offline Reinforcement Learning. In *International Conference on Learning Representations (ICLR)*. arXiv:2010.13611.
- [2] Kevin Black, Noah Brown, James Darpanian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haohuan Wang, Lili Yu, and Ury Zhilinsky. 2025. $\pi_{0.5}$: A Vision-Language-Action Model with Open-World Generalization. *arXiv preprint* (2025). Physical Intelligence. arXiv:2504.16054.
- [3] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. 2024. π_0 : A Vision-Language-Action Flow Model for General Robot Control. *arXiv preprint* (2024). Physical Intelligence. arXiv:2410.24164.
- [4] Konstantinos Bousmalis, Giulia Vezzani, Dushyant Rao, Coline Devin, Alex X. Lee, Maria Bauza, et al. 2023. RoboCat: A Self-Improving Generalist Agent for Robotic Manipulation. *Transactions on Machine Learning Research (TMLR)* (2023). arXiv:2306.11706.
- [5] Anthony Brohan, Noah Brown, Justice Carbajal, et al. 2023. RT-1: Robotics Transformer for Real-World Control at Scale. In *Robotics: Science and Systems (RSS)*. arXiv:2212.06817.
- [6] Anthony Brohan, Noah Brown, Justice Carbajal, et al. 2023. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In *Conference on Robot Learning (CoRL)*. arXiv:2307.15818.
- [7] Harold Haodong Chen, Sirui Chen, Yingjie Xu, Wenhong Ge, and Ying-Cong Chen. 2026. RoboEvolve: Co-Evolving Planner-Simulator for Robotic Manipulation with Limited Data. *arXiv preprint* (2026). arXiv:2605.13775.
- [8] Junting Chen, Yao Mu, Qiaojun Yu, Tianming Wei, Silang Wu, Zhecheng Yuan, Zhixuan Liang, Chao Yang, Kaipeng Zhang, Wenqi Shao, Yu Qiao, Huazhe Xu, Mingyu Ding, and Ping Luo. 2024. RoboScript: Code Generation for Free-Form Manipulation Tasks across Real and Simulation. *arXiv preprint* (2024). arXiv:2402.14623.
- [9] Lawrence Yunliang Chen, Kush Hari, Karthik Dharmarajan, Chenfeng Xu, Quan Vuong, and Ken Goldberg. 2024. Mirage: Cross-Embodiment Zero-Shot Policy Transfer with Cross-Painting. In *Robotics: Science and Systems (RSS)*. arXiv:2402.19249.
- [10] Ria Doshi, Homer Walke, Oier Mees, Sudeep Dasari, and Sergey Levine. 2024. Scaling Cross-Embodied Learning: One Policy for Manipulation, Navigation, Locomotion and Aviation. In *Conference on Robot Learning (CoRL)*. arXiv:2408.11812.
- [11] Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. 2024. AHA: A Vision-Language-Model for Detecting and Reasoning Over Failures in

- Robotic Manipulation. *arXiv preprint* (2024). arXiv:2410.00371.
- [12] Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine. 2019. Diversity is All You Need: Learning Skills without a Reward Function. In *International Conference on Learning Representations (ICLR)*. arXiv:1802.06070.
 - [13] Letian Fu, Justin Yu, Karim El-Refaï, Ethan Kou, Haoru Xue, Huang Huang, Wenli Xiao, Guanzhi Wang, Dantong Niu, Li Fei-Fei, Guanya Shi, Jiajun Wu, Shankar Sastry, Yuke Zhu, Ken Goldberg, and Linxi Fan. 2026. CaP-X: A Framework for Benchmarking and Improving Coding Agents for Robot Manipulation. *arXiv preprint* (2026). arXiv:2603.22435.
 - [14] Gemini Robotics Team, Google DeepMind. 2025. Gemini Robotics: Bringing AI into the Physical World. *arXiv preprint* (2025). arXiv:2503.20020.
 - [15] Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Pannag Sanketi, Quan Vuong, Ted Xiao, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. 2024. Octo: An Open-Source Generalist Robot Policy. In *Robotics: Science and Systems (RSS)*. Octo Model Team. arXiv:2405.12213.
 - [16] Jiayuan Gu, Fanbo Xiang, Xuanlin Li, Zhan Ling, Xiqiang Liu, Tongzhou Mu, Yihe Tang, Stone Tao, Xinyue Wei, Yunchao Yao, Xiaodi Yuan, Pengwei Xie, Zhiao Huang, Rui Chen, and Hao Su. 2023. ManiSkill2: A Unified Benchmark for Generalizable Manipulation Skills. In *International Conference on Learning Representations (ICLR)*. arXiv:2302.04659.
 - [17] Ping Guo, Chao Li, Yinglan Feng, and Chaoning Zhang. 2026. Code Evolution for Control: Synthesizing Policies via LLM-Driven Evolutionary Search. *arXiv preprint* (2026). arXiv:2601.06845.
 - [18] Yanjiang Guo, Yen-Jen Wang, Lihan Zha, and Jianyu Chen. 2024. DoReMi: Grounding Language Model by Detecting and Recovering from Plan-Execution Misalignment. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. arXiv:2307.00329.
 - [19] Siyuan Huang, Zhengkai Jiang, Hao Dong, Yu Qiao, Peng Gao, and Hongsheng Li. 2023. Instruct2Act: Mapping Multi-modality Instructions to Robotic Actions with Large Language Model. *arXiv preprint* (2023). arXiv:2305.11176.
 - [20] Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. 2023. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models. In *Conference on Robot Learning (CoRL)*. arXiv:2307.05973.
 - [21] Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Noah Brown, Tomas Jackson, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. 2022. Inner Monologue: Embodied Reasoning through Planning with Language Models. In *Conference on Robot Learning (CoRL)*. arXiv:2207.05608.
 - [22] Stephen James, Zicong Ma, David Rovick Arrojo, and Andrew J. Davison. 2020. RLBech: The Robot Learning Benchmark and Learning Environment. *IEEE Robotics and Automation Letters (RA-L)* (2020). arXiv:1909.12271.
 - [23] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. 2024. OpenVLA: An Open-Source Vision-Language-Action Model. In *Conference on Robot Learning (CoRL)*. arXiv:2406.09246.
 - [24] Michael Laskin, Hao Liu, Xue Bin Peng, Denis Yarats, Aravind Rajeswaran, and Pieter Abbeel. 2022. CIC: Contrastive Intrinsic Control for Unsupervised Skill Discovery. *arXiv preprint* (2022). arXiv:2202.00161.
 - [25] Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, et al. 2024. BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation. *arXiv preprint* (2024). arXiv:2403.09227.
 - [26] Hao Li, Xue Yang, Zhaokai Wang, Xizhou Zhu, Jie Zhou, Yu Qiao, Xiaogang Wang, Hongsheng Li, Lewei Lu, and Jifeng Dai. 2024. Auto MC-Reward: Automated Dense Reward Design with Large Language Models for Minecraft. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. arXiv:2312.09238.
 - [27] Jingyao Li, Pengguang Chen, Sitong Wu, Chuanyang Zheng, Hong Xu, and Jiaya Jia. 2024. RoboCoder: Robotic Learning from Basic Skills to General Tasks with Large Language Models. *arXiv preprint* (2024). arXiv:2406.03757.
 - [28] Qixiu Li, Yaobo Liang, Zeyu Wang, Lin Luo, Xi Chen, Mozheng Liao, Fangyun Wei, Yu Deng, Sicheng Xu, Yizhong Zhang, Xiaofan Wang, Bei Liu, Jianlong Fu, Jianmin Bao, Dong Chen, Yuanchun Shi, Jiaolong Yang, and Baining Guo. 2024. CogACT: A Foundational Vision-Language-Action Model for Synergizing Cognition and Action in Robotic Manipulation. *arXiv preprint* (2024). arXiv:2411.19650.
 - [29] Ruiying Li, Yunlang Zhou, YuYao Zhu, Kylin Chen, Jingyuan Wang, Sukai Wang, Kongtao Hu, Minhui Yu, Bowen Jiang, Zhan Su, Jiayao Ma, Xin He, Yongjian Shen, Yang Yang, Guanghui Ren, Maoqing Yao, Wenhao Wang, and Yao Mu. 2026. RoboClaw: An Agentic Framework for Scalable Long-Horizon Robotic Tasks. *arXiv preprint* (2026). arXiv:2603.11558.
 - [30] Xinghang Li, Minghuan Liu, Hanbo Zhang, Cunjun Yu, Jie Xu, Hongtao Wu, Chilam Cheang, Ya Jing, Weinan Zhang, Huaping Liu, Hang Li, and Tao Kong. 2024. Vision-Language Foundation Models as Effective Robot Imitators. In *International Conference on Learning Representations (ICLR)*. RoboFlamingo. arXiv:2311.01378.
 - [31] Zaijing Li, Yuquan Xie, Rui Shao, Gongwei Chen, Dongmei Jiang, and Liqiang Nie. 2024. Optimus-1: Hybrid Multimodal Memory Empowered Agents Excel in Long-Horizon Tasks. In *Advances in Neural Information Processing Systems*

- (*NeurIPS*). arXiv:2408.03615.
- [32] Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. 2023. Code as Policies: Language Model Programs for Embodied Control. In *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2209.07753.
 - [33] Kaiqu Liang, Zixu Zhang, and Jaime Fernández Fisac. 2024. Introspective Planning: Aligning Robots' Uncertainty with Inherent Task Ambiguity. *arXiv preprint* (2024). arXiv:2402.06529.
 - [34] William Liang, Sam Wang, Hung-Ju Wang, Osbert Bastani, Dinesh Jayaraman, and Yecheng Jason Ma. 2024. Eureka-verse: Environment Curriculum Generation via Large Language Models. In *Conference on Robot Learning (CoRL)*. arXiv:2411.01775.
 - [35] Kevin Lin, Christopher Agia, Toki Migimatsu, Marco Pavone, and Jeannette Bohg. 2023. Text2Motion: From Natural Language Instructions to Feasible Plans. *Autonomous Robots* (2023). arXiv:2303.12153.
 - [36] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. 2023. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. arXiv:2306.03310.
 - [37] Shunyu Liu, Yaoru Li, Kongcheng Zhang, Zhenyu Cui, Wenkai Fang, Yuxuan Zheng, Tongya Zheng, and Mingli Song. 2024. Odyssey: Empowering Minecraft Agents with Open-World Skills. *arXiv preprint* (2024). arXiv:2407.15325.
 - [38] Zeyi Liu, Arpit Bahety, and Shuran Song. 2023. REFLECT: Summarizing Robot Experiences for Failure Explanation and Correction. In *Conference on Robot Learning (CoRL)*. arXiv:2306.15724.
 - [39] Runyu Lu, Yubo Wu, Ethan Kou, Letian Fu, Wenli Xiao, Ajay Mandlekar, Yinzen Xu, Guanya Shi, Ken Goldberg, Ang Chen, Mosharaf Chowdhury, Yuke Zhu, Linxi Fan, and Guanzhi Wang. 2026. ASPIRE: Agentic Skills Discovery for Robotics. *arXiv preprint* (2026). arXiv:2607.00272.
 - [40] Yecheng Jason Ma, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh Jayaraman, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024. Eureka: Human-Level Reward Design via Coding Large Language Models. In *International Conference on Learning Representations (ICLR)*. arXiv:2310.12931.
 - [41] Yecheng Jason Ma, William Liang, Hung-Ju Wang, Sam Wang, Yuke Zhu, Linxi Fan, Osbert Bastani, and Dinesh Jayaraman. 2024. DrEureka: Language Model Guided Sim-To-Real Transfer. In *Robotics: Science and Systems (RSS)*. arXiv:2406.01967.
 - [42] Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. 2022. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks. *IEEE Robotics and Automation Letters (RA-L)* (2022). arXiv:2112.03227.
 - [43] Yao Mu, Junting Chen, Qinglong Zhang, Shoufa Chen, Qiaojun Yu, Chongjian Ge, Runjian Chen, Zhixuan Liang, Mengkang Hu, Chaofan Tao, Peize Sun, Haibao Yu, Chao Yang, Wenqi Shao, Wenhai Wang, Jifeng Dai, Yu Qiao, Mingyu Ding, and Ping Luo. 2024. RoboCodeX: Multimodal Code Generation for Robotic Behavior Synthesis. In *International Conference on Machine Learning (ICML)*. arXiv:2402.16117.
 - [44] NVIDIA, Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, et al. 2025. GR00T N1: An Open Foundation Model for Generalist Humanoid Robots. *arXiv preprint* (2025). NVIDIA. arXiv:2503.14734.
 - [45] Open X-Embodiment Collaboration. 2024. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. In *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2310.08864.
 - [46] Seohong Park, Jongwook Choi, Jaekyeom Kim, Honglak Lee, and Gunhee Kim. 2022. Lipschitz-constrained Unsupervised Skill Discovery. In *International Conference on Learning Representations (ICLR)*. arXiv:2202.00914.
 - [47] Seohong Park, Oleh Rybkin, and Sergey Levine. 2024. METRA: Scalable Unsupervised RL with Metric-Aware Abstraction. In *International Conference on Learning Representations (ICLR)*. arXiv:2310.08887.
 - [48] Karl Pertsch, Youngwoon Lee, and Joseph J. Lim. 2020. Accelerating Reinforcement Learning with Learned Skill Priors. In *Conference on Robot Learning (CoRL)*. arXiv:2010.11944.
 - [49] Delin Qu, Haoming Song, Qizhi Chen, Yuanqi Yao, Xinyi Ye, Yan Ding, Zhigang Wang, JiaYuan Gu, Bin Zhao, Dong Wang, and Xuelong Li. 2025. SpatialVLA: Exploring Spatial Representations for Visual-Language-Action Model. *arXiv preprint* (2025). arXiv:2501.15830.
 - [50] Archit Sharma, Shixiang Gu, Sergey Levine, Vikash Kumar, and Karol Hausman. 2020. Dynamics-Aware Unsupervised Discovery of Skills. In *International Conference on Learning Representations (ICLR)*. arXiv:1907.01657.
 - [51] Lucy Xiaoyang Shi, Joseph J. Lim, and Youngwoon Lee. 2022. Skill-based Model-based Reinforcement Learning. In *Conference on Robot Learning (CoRL)*. arXiv:2207.07560.
 - [52] Avi Singh, Huihan Liu, Gaoyue Zhou, Albert Yu, Nicholas Rhinehart, and Sergey Levine. 2021. Parrot: Data-Driven Behavioral Priors for Reinforcement Learning. In *International Conference on Learning Representations (ICLR)*. arXiv:2011.10024.
 - [53] Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. 2023. ProgPrompt: Generating Situated Robot Task Plans using Large Language Models. In *IEEE*

- International Conference on Robotics and Automation (ICRA)*. arXiv:2209.11302.
- [54] Georgios Tzafas and Hamidreza Kasaei. 2024. Lifelong Robot Library Learning: Bootstrapping Composable and Generalizable Skills for Embodied Control with Language Models. In *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2406.18746.
 - [55] Unitree Robotics. 2026. UniStore: A Humanoid Robot Application Store. <https://unistore.unitree.com>.
 - [56] Sai Vemprala, Rogerio Bonatti, Arthur Buckner, and Ashish Kapoor. 2023. ChatGPT for Robotics: Design Principles and Model Abilities. *arXiv preprint* (2023). arXiv:2306.17582.
 - [57] Weikang Wan, Yifeng Zhu, Rutav Shah, and Yuke Zhu. 2024. LOTUS: Continual Imitation Learning for Robot Manipulation Through Unsupervised Skill Discovery. In *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2311.02058.
 - [58] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023. Voyager: An Open-Ended Embodied Agent with Large Language Models. *arXiv preprint* (2023). arXiv:2305.16291.
 - [59] Huaxiaoyue Wang, Gonzalo Gonzalez-Pumariega, Yash Sharma, and Sanjiban Choudhury. 2023. Demo2Code: From Summarizing Demonstrations to Synthesizing Code via Extended Chain-of-Thought. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2305.16744.
 - [60] Yufei Wang, Zhou Xian, Feng Chen, Tsun-Hsuan Wang, Yian Wang, Katerina Fragkiadaki, Zackory Erickson, David Held, and Chuang Gan. 2024. RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation. In *International Conference on Machine Learning (ICML)*. arXiv:2311.01455.
 - [61] Yen-Jen Wang, Bike Zhang, Jianyu Chen, and Koushil Sreenath. 2024. Prompt a Robot to Walk with Large Language Models. In *IEEE Conference on Decision and Control (CDC)*. arXiv:2309.09969.
 - [62] Zihao Wang, Shaofei Cai, Anji Liu, Yonggang Jin, Jinbing Hou, Bowei Zhang, Haowei Lin, Zhaofeng He, Zilong Zheng, Yaodong Yang, Xiaojian Ma, and Yitao Liang. 2023. JARVIS-1: Open-World Multi-task Agents with Memory-Augmented Multimodal Language Models. *arXiv preprint* (2023). arXiv:2311.05997.
 - [63] Jimmy Wu, Rika Antonova, Adam Kan, Marion Lepert, Andy Zeng, Shuran Song, Jeannette Bohg, Szymon Rusinkiewicz, and Thomas Funkhouser. 2023. TidyBot: Personalized Robot Assistance with Large Language Models. *Autonomous Robots* (2023). arXiv:2305.05658.
 - [64] Wenli Xiao, Jia Xie, Tonghe Zhang, Haotian Lin, Letian Fu, Haoru Xue, Jalen Lu, Yi Yang, Cunxi Dai, Zi Wang, Jimmy Wu, Guanzhi Wang, S. Shankar Sastry, Ken Goldberg, Linxi Fan, Yuke Zhu, and Guanya Shi. 2026. ENPIRE: Agentic Robot Policy Self-Improvement in the Real World. *arXiv preprint* (2026). arXiv:2606.19980.
 - [65] Senwei Xie, Hongyu Wang, Zhanqi Xiao, Ruiping Wang, and Xilin Chen. 2025. Robotic Programmer: Video Instructed Policy Code Generation for Robotic Manipulation. *arXiv preprint* (2025). arXiv:2501.04268.
 - [66] Senwei Xie, Yuntian Zhang, Ruiping Wang, and Xilin Chen. 2026. Uni-Skill: Building Self-Evolving Skill Repository for Generalizable Robotic Manipulation. In *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2603.02623.
 - [67] Tianbao Xie, Siheng Zhao, Chen Henry Wu, Yitao Liu, Qian Luo, Victor Zhong, Yanchao Yang, and Tao Yu. 2024. Text2Reward: Reward Shaping with Language Models for Reinforcement Learning. In *International Conference on Learning Representations (ICLR)*. arXiv:2309.11489.
 - [68] Takuma Yoneda, Jiading Fang, Peng Li, Huanyu Zhang, Tianchong Jiang, Shengjie Lin, Ben Picker, David Yunis, Hongyuan Mei, and Matthew R. Walter. 2024. Statler: State-Maintaining Language Models for Embodied Reasoning. In *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2306.17840.
 - [69] Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Avnish Narayan, Hayden Shively, Adithya Bellathur, Karol Hausman, Chelsea Finn, and Sergey Levine. 2019. Meta-World: A Benchmark and Evaluation for Multi-Task and Meta Reinforcement Learning. In *Conference on Robot Learning (CoRL)*. arXiv:1910.10897.
 - [70] Wenhao Yu, Nimrod Gileadi, Chuyuan Fu, Sean Kirmani, Kuang-Huei Lee, Montse Gonzalez Arenas, Hao-Tien Lewis Chiang, Tom Erez, Leonard Hasenclever, Jan Humplik, Brian Ichter, Ted Xiao, Peng Xu, Andy Zeng, Tingnan Zhang, Nicolas Heess, Dorsa Sadigh, Jie Tan, Yuval Tassa, and Fei Xia. 2023. Language to Rewards for Robotic Skill Synthesis. In *Conference on Robot Learning (CoRL)*. arXiv:2306.08647.
 - [71] Jesse Zhang, Karl Pertsch, Jiahui Zhang, and Joseph J. Lim. 2024. SPRINT: Scalable Policy Pre-Training via Language Instruction Relabeling. In *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2306.11886.
 - [72] Jesse Zhang, Jiahui Zhang, Karl Pertsch, Ziyi Liu, Xiang Ren, Minsuk Chang, Shao-Hua Sun, and Joseph J. Lim. 2023. Bootstrap Your Own Skills: Learning to Solve New Tasks with Large Language Model Guidance. In *Conference on Robot Learning (CoRL)*. arXiv:2310.10021.
 - [73] Ziao Zhang, Kou Shi, Shiting Huang, Avery Nie, Yu Zeng, Yiming Zhao, Zhen Fang, Qisheng Su, Haibo Qiu, Wei Yang, Qingnan Ren, Shun Zou, Wenxuan Huang, Lin Chen, Zehui Chen, and Feng Zhao. 2026. SkillFlow: Benchmarking Lifelong Skill Discovery and Evolution for Autonomous Agents. *arXiv preprint* (2026). arXiv:2604.17308.
 - [74] Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. ExpeL: LLM Agents Are Experiential Learners. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. arXiv:2308.10144.

- [75] Enshen Zhou, Qi Su, Cheng Chi, Zhizheng Zhang, Zhongyuan Wang, Tiejun Huang, Lu Sheng, and He Wang. 2025. Code-as-Monitor: Constraint-aware Visual Programming for Reactive and Proactive Robotic Failure Detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. arXiv:2412.04455.
- [76] Xizhou Zhu, Yuntao Chen, Hao Tian, Chenxin Tao, Weijie Su, Chenyu Yang, Gao Huang, Bin Li, Lewei Lu, Xiaogang Wang, Yu Qiao, Zhaoxiang Zhang, and Jifeng Dai. 2023. Ghost in the Minecraft: Generally Capable Agents for Open-World Environments via Large Language Models with Text-based Knowledge and Memory. *arXiv preprint* (2023). arXiv:2305.17144.
- [77] Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Kevin Lin, Abhiram Maddukuri, Soroush Nasiriany, and Yifeng Zhu. 2020. robosuite: A Modular Simulation Framework and Benchmark for Robot Learning. *arXiv preprint* (2020). arXiv:2009.12293.